Prototype Design for Rubik's Cube Solver

A. M. Andrew, W. Faridah, W. H. Tan, S. Ragunathan, A. S. N. Amirah, N. A. N. Zainab, and F. S. Lee

Abstract Rubik's cube is a modern day plastic material puzzle, where the one need to twist and turn to solve the multicolour squares. At the end, the puzzle need to be solved to make all the same colour cubes on the same surfaces. High intelligence is needed to solve it at a shorter time. Therefore, this research is to create a robot solver to solve the Rubik's cube in short time smartly. Initially, the algorithm will scan the six surfaces of unsolved cube through a webcam and register the colours in the memory. Once it is completed, the solving algorithm will analyse and instruct the servo motors to twist and turn based on the computed solution. It is then pass it to the solving algorithm to identify the solving process and send the moving instruction to the motors by Raspberry Pi. The robot body was constructed using laser cutter. It is to make sure that the measurements are accurate and correct. To solve the cube, four servo motors with high efficiency were used to twist the cube in certain legit with the rules of the game. Another four servo motors are connected to the rack and pinion gears by clamping to hold the Rubik's cube from falling when it is twisted. An improvised "Kociemba Algorithm", also called as "The Two-Phase-Algorithm" is used. It reduced the required moves to solve the cube into a maximum of 25 moves and a minimum of 19. Conclusively, the final prototype discussed in this paper is tested with the solving algorithm. The results are presented.

Keywords Rubik's cube · Robot solver · Image processing · Raspberry · Pi · Servo motor

A. M. Andrew (✉) · F. S. Lee
Faculty of Electrical Engineering Technology, Universiti Malaysia Perlis (UniMAP), Pauh Putra Campus, 02600 Arau, Perlis, Malaysia
e-mail: allanmelvin@unimap.edu.my

W. Faridah · S. Ragunathan · A. S. N. Amirah · N. A. N. Zainab
Faculty of Civil Engineering Technology, Universiti Malaysia Perlis (UniMAP), Kompleks Pusat Pengajian Jejawi 3, 02600 Arau, Perlis, Malaysia

W. H. Tan
Faculty of Mechanical Engineering Technology, Universiti Malaysia Perlis (UniMAP), Pauh Putra Campus, 02600 Arau, Perlis, Malaysia

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1 Introduction

The Rubik's cube has been one of the most interesting and challenging puzzles ever built. It continues to remain in the top among the puzzle games because of the design of the puzzle [1]. For instance, one correct solution is out of the 43 quintillion other possibilities [2]. If a solution needs 1 byte, the total memory needed will be 43 million terabytes hard disks size. In construction, the Rubik's cube is a cube structure with $3 \times 3 \times 3$ smaller cubes arranged in it. It has a total of six colours on its six faces. All the smaller cubes arranged around it can be moved vertically or horizontally. A professional solver can solve the cube as fast as less than one minute, and an amateur solver can take even hours to solve it [3]. There are other varieties and sizes of Rubik's cube available, with $33 \times 33 \times 33$ size is the solvable cube with Guinness World Record [2].

However, only those who know the trick can solve the shuffled cube. McManus, E. (2018) mentioned that the mathematical formations behind Rubik's cube make it solvable [2]. It might take minutes, hours or days to solve it. It starts from a step-by-step solving of the scrambled cube's layer until each surface have same colour. Therefore, having intelligent algorithm with image processing capability implemented and Internet-of-Things (IoT) in designed mechanical structure, a robot could be made to solve it [4]. A good mechanical design must be supported by the right algorithm [5, 6]. Thus, in this research, a cage-like prototype is constructed to hold and solve the Rubik's cube. The developed robotic system can solve any randomised 3×3 Rubik's cube with less than 25 steps of rotations.

The main objective of this research is to build a prototype to support the Rubik's cube solver algorithm that is able to read the initial state of the Rubik's cube reliably, find a solution and then solve the cube using the solver.

2 Literature Review

There are several methods and devices used to solve the Rubik's cube [7]. Several researches had performed by many institutions throughout the world to offer the best Rubik's cube Solver [2]. Therefore, this section gives a brief survey on various Rubik's Cube solvers in the world.

2.1 Mind Cuber [2]

MindCuber is a single armed solver by David Gilday. The single arm is responsible for holding the cube in place whilst the lower platform rotates the D face of the cube. The single arm is also responsible for performing cube rotations. This robot can be built using an EV3 LEGO Mindstorms set which makes it cheaper to be built. The

limitations lie with its design. Since only a single side can be turned at a time, the cube needs to be rotated every time a face need to be turned. This is extremely time consuming. The MindCuber uses a single Red, Green and Blue (RGB) sensor to read each square individually [8].

2.2 JPBrown's CubeSolver [2]

JPBrown's CubeSolver is one of the first attempt in building a cube solving robot. The robot uses 3 arms built from LEGO. This allows it to move three independent faces without cube rotations. JPBrown's clamping mechanism uses a complex gearing system which makes the face move slowly but accurately [9, 10]. The vision system is webcam based. The cube must be presented in a very specific area of the camera frame. The robot itself is powered by 2 Retrocommissioning (RCX) Intelligent Bricks and a Personal Computer (PC). The PC is responsible for finding a solution and parsing the camera frames for the vision. The Bricks are responsible for robot movement. Since the CubeSolver has a PC at its disposal, Kociemba's algorithm is used as the algorithm of choice. This is because it will find a solution within a relatively short time. Since the execution of moves is slow, the solution needs to be as short as possible.

2.3 GoCube [2]

GoCube is a new and improved puzzle that is fitted with smart sensors and Bluetooth connectivity to make puzzle-solving more engaging. It is basically designed to track and measure the movements in order to help players to solve the classic puzzle, sharpen the skills and beat other players too. Moreover, it is built as a speed cube. It is designed with special mechanisms to make it turn smoother, faster and less clunky than the classic Rubik's cube, making it more enjoyable to play with. This smart connected toy can also help players to learn how to solve the puzzle in just one hour. From there, the player can keep on tracking and improving their solving skills to solve the cube in shorter time.

2.4 MIT Rubik's Cube Solver [2]

The Rubik's cube solver built by MIT can solve the 3×3 cube within a second, with staggering 20 moves/seconds. The project was targeted as the fastest Rubik's Cube Solver in MIT. The project used six stepper motors as actuators. However, this project uses no sensor. The middle Rubik's cube could not be taken out as is it fixed to the main body since they only focused on fast solving.

3 Methodology

In this section, the methodology used in this developing the prototype design is carefully explained. Figure 1 shows the flowchart of the prototype design. Before building the 3 × 3 Rubik's cube solver prototype, the required components are identified. When the fabrication of the solver robot prototype has been completed, a few test runs are performed to make sure that the prototype is functioning properly.

The frame of the robot is carefully designed in Fusion 360 designing software. The Computer-Aided Design (CAD) was developed around the size of the cube. The design is meant to be easy to be manufactured, simple to construct and repair, and most importantly, providing a relatively unobstructed view of the cube during the entire solving process [9, 10]. Once it is perfected in the software, the design is transferred to laser cutting machine for physical prototype preparation. 5 and 10 mm acrylic are used for the manufacturing process. Figure 2 shows the base design using Fusion 360.

Figure 3 shows the base prototype done by laser cutting process. The prototype is to be rigid and strong, so that, the movements of the servo motors can be translated into precise rotations of the Rubik's cube faces. The robot is designed to consist of only two unique parts, consists of only two identical plates and four identical side plates. Having identical parts enables to create more spare parts in case of replacement needed. The main idea of having minimalist and transparent materials in the prototype design is to provide unobstructed view during the cube solving process.

3.1 Rack and Pinion

Rack and pinions are the type of mechanical gears which enables the rotary to translational movement conversion or vice versa. In this prototype, the rack is attached to servo motor MG92B. The height has been set to ensure the Rubik's cube can be twisted fully 360°. On the other hand, the pinion is attached to servo motor SG92R. The pinion will be static. When the servo motor SG92R turns, the rack will follow it.

The rack and pinions are fixed in rigid condition. Figures 4 and 5 shows the rack and pinion, designed in Fusion 360 and printed in 3D printer, respectively. The 3D printed parts are connected to the base and frames using a small sized screws and nuts.

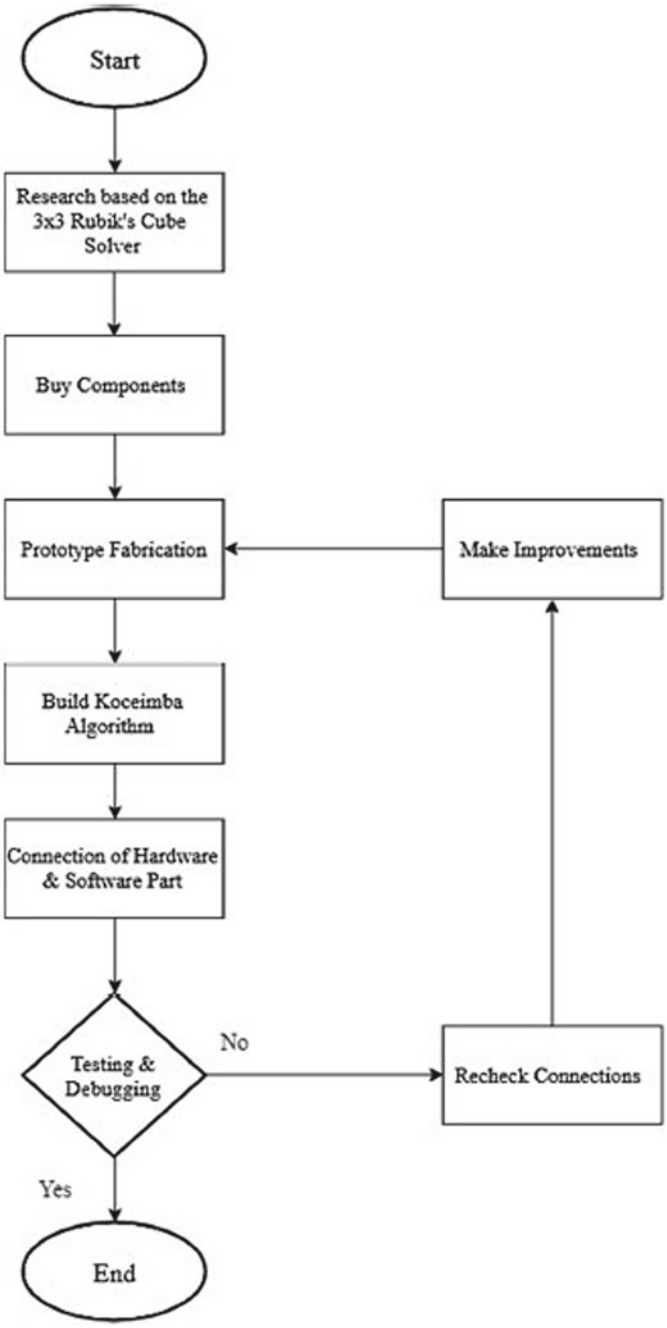

Fig. 1 Prototype design flowchart

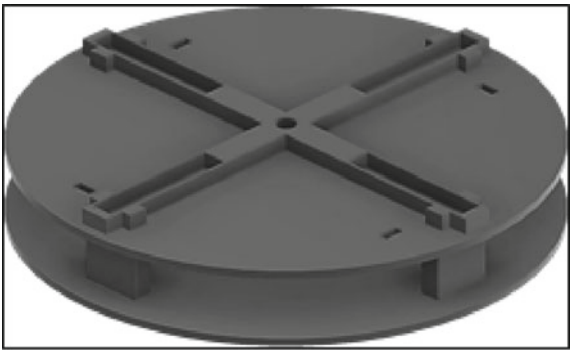

Fig. 2 Base design using fusion 360

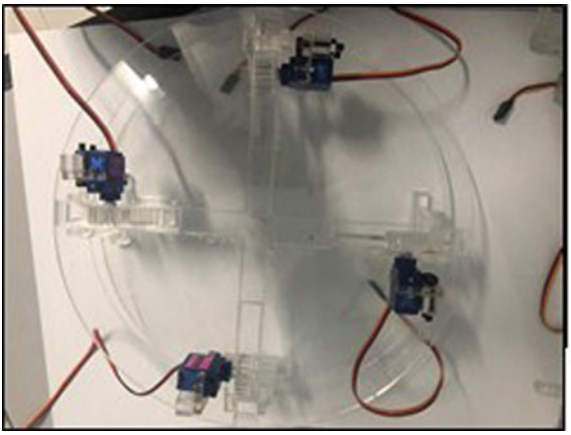

Fig. 3 Base prototype using laser cutting

3.2 Logitech C270 Webcam

In this research, Logitech C270 webcam is used to capture the six faces of the scrambled cubes. The captured images are used as the inputs for the image processing algorithm. The webcam is controlled by Raspberry Pi module. Webcam provides more accurate and fast input compared to other vision cameras. Webcams also are easily available and cost effective compared to other vision sensors. Webcam will convert colour signal into binary format and arrange it in matrix order. This will allow computer to process image quickly using image processing. MATLAB and Python is used to convert the images to colour input. Figure 6 shows the webcam used in this research.

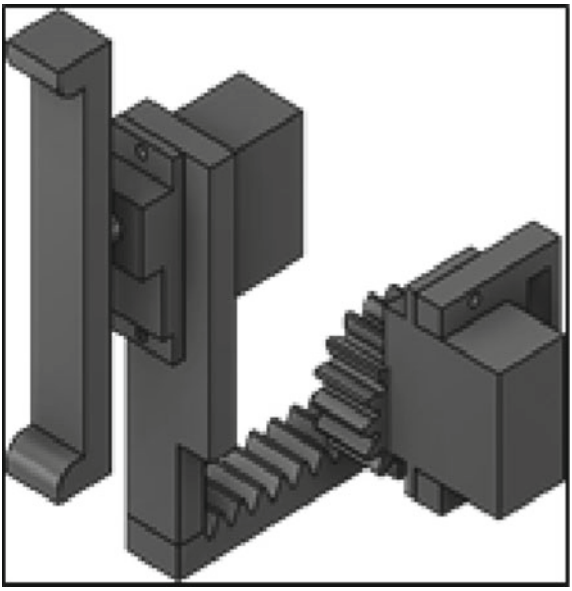

Fig. 4 Rack and Pinion using fusion 360

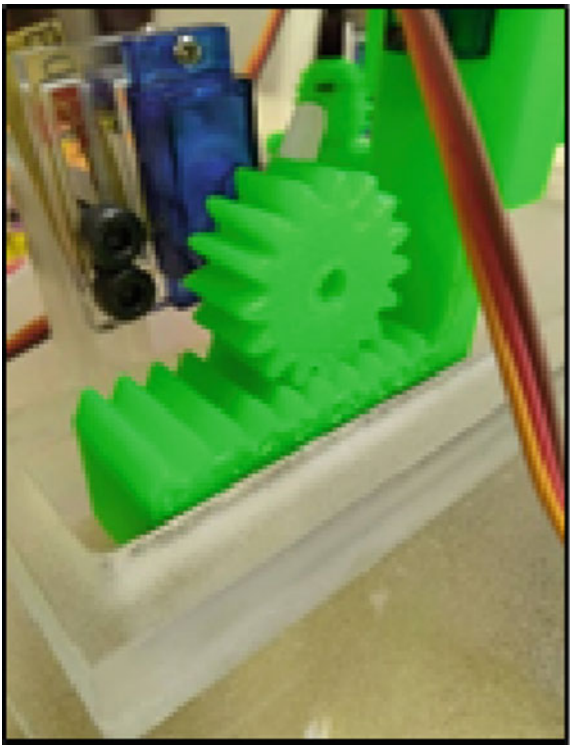

Fig. 5 Rack and Pinion by 3D printer

Fig. 6 Logitech C270 webcam used in this prototype

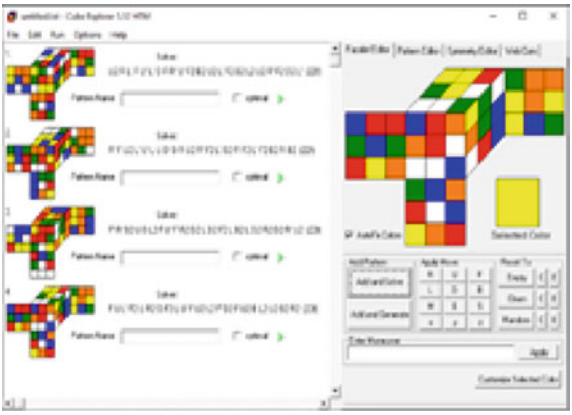

Fig. 7 Thistlethwaite's algorithm used in cube explorer software

3.3 Kociemba Algorithm

Kociemba's algorithm is a two-phase algorithm that is also known as God's algorithm. God's algorithm is a fancy term of most efficient algorithm in the sense that the Rubik's Cube can be turned in the fewest number of times to be solved in any given state. Kociemba's algorithm allows for the Rubik's cube to be solved in only less than 25 turns. Kociemba's algorithms is considered a twophase algorithm because after an initial solution is reached, additional optimal solutions are searched for solving. It was developed from the former most efficient algorithm known as Thistlethwaite's algorithm. It solves the cube by looking at it in groups and phases like in Fig. 7. The

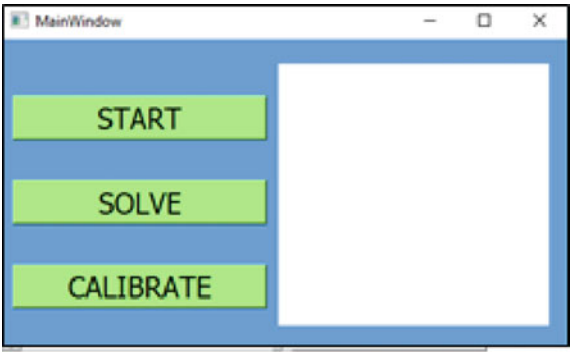

Fig. 8 Graphical user interface developed using phyton

first group is the G0 group which is the randomly mixed cube group. At this stage there are no restrictions on what moves can be made when utilizing this algorithm. To complete this state, the cube needs at most 12 moves. The next group is the G1 group which is when the cube gets in a specific state of U, D, L2, R2, F2, B2. When the cube is in this state, there are specific moves that cannot be made. To complete this state, the cube needs at most 25 moves. This method is used in this research. Figure 8 shows the Graphical User Interface developed for this research using Phyton language.

3.4 Raspberry Pi 3B+

Raspberry Pi 3B+ is used in this research to store and process data. The motor is controlled by Raspberry Pi through General-Purpose Input/Output (GPIO) pin. This Raspberry Pi module uses 1.4 GHz microprocessor and 1 GB RAM which allows the image processing to run smoothly. This project uses Virtual Network Computing (VNC) which allows the project to run from a distance without the need of any physical connection.

Figure 9 shows the complete prototype design of the Rubik's cube solver developed using Fusion 360 design software in this research. Figure 10 shows the physical prototype made from acrylic and 3D printing materials.

4 Experimental Result and Discussions

The final prototype is tested with the solving algorithm developed using Phyton language. The cube managed to be solved successfully in 19 moves. Figure 11 shows the Phyton programming and the developed GUI used for solving the cube. The program has 3 functions, Start, Solve and Calibrate.

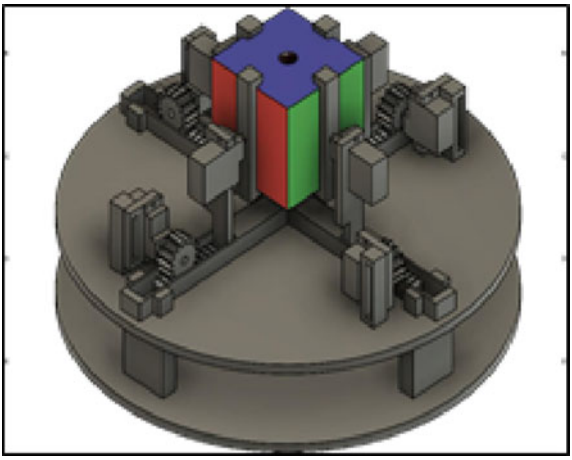

Fig. 9 Complete prototype design using fusion 360 software

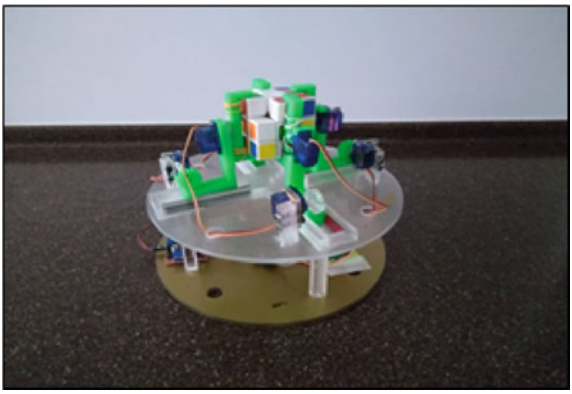

Fig. 10 Complete prototype using acrylic and 3D printer material

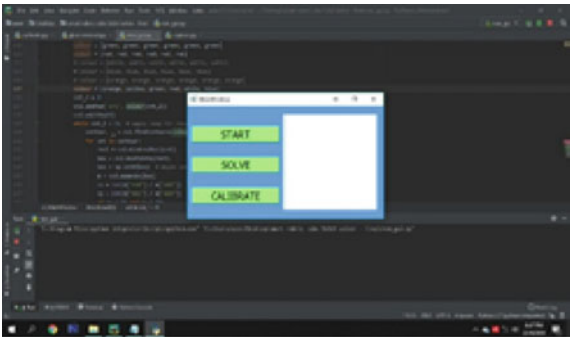

Fig. 11 Testing with Kociemba algorithm for Rubik's cube solving

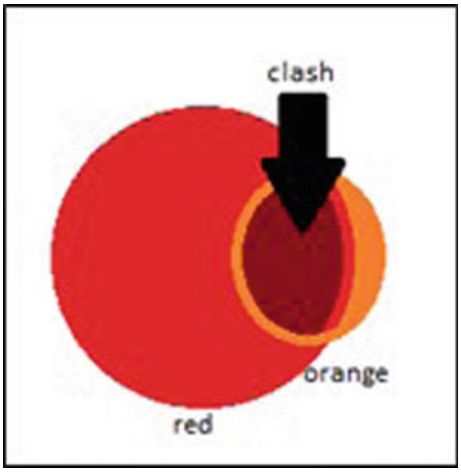

Fig. 12 Colour space redundancy between red and orange colour

START – to capture six images.
SOLVE – to calculate solution and to actuate servo for Rubik's cube solving.
CALIBRATE – to set boundaries of the hue, saturation and value for colour contours in system.

The program works with 97% average accuracy, tested in various lighting conditions. The faces of Rubik's cube can be capture easily when a proper lighting condition present. However, for some cases, there is misclassification between red and orange colour due to almost similar colour shades, which makes orange colour to be classified as red. Figure 12 shows the colour space redundancy between red and orange colours. The range of orange colour is overlapping with red. In order to overcome this misclassification, the algorithm is improvised with the correction algorithm. Firstly, the program detects the red pixels (Step 1). There is a chance of few orange pixels to be misclassified as red. Then, the orange pixels are detected separately (Step 2). Step 1 is compared to Step 2, and the overlapping pixels are identified. The overlapping pixels are re-registered as orange pixels.

5 Conclusion

In conclusion, the paper focuses on the method involved in building the Rubik's cube solver prototype, and on how this research combines the prototype and the algorithm to solve the Rubik's cube successfully. The prototype uses 8 servo motors that are controlled by one pin I2C from raspberry PI. The base of the prototype is made of acrylic and 3D printing materials. The prototype is designed in such a way that the view of solving the cube will not be obstructed by components. The images captured using webcam and the images are processed by Raspberry Pi to create solution for the Rubik's cube. Phyton programming language is used for programming of the

Kociemba algorithm used in solving the Rubik's cube. This research comprises the methods and experimental results in development of software for solving Rubik's cube, no matter how hard the combination is. The main task of the program to create a solution for any scrambled Rubik's cube. The prototype and the solving program can be used and improved further for education purpose. In the future research, the unfilled research gaps can be filled. The usage of better materials and motors can increase the efficiency of the robot immensely.

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